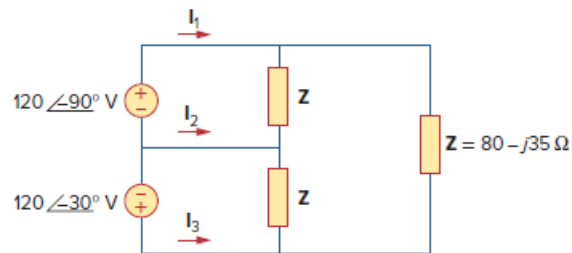


Problem

Use mesh analysis to find currents I_1 , I_2 , and I_3 in the circuit of Fig. 10.82.

Figure 10.82:



Step-by-step solution

Step 1 of 4

Refer to Figure 10.82 in the textbook.

Redraw Figure,

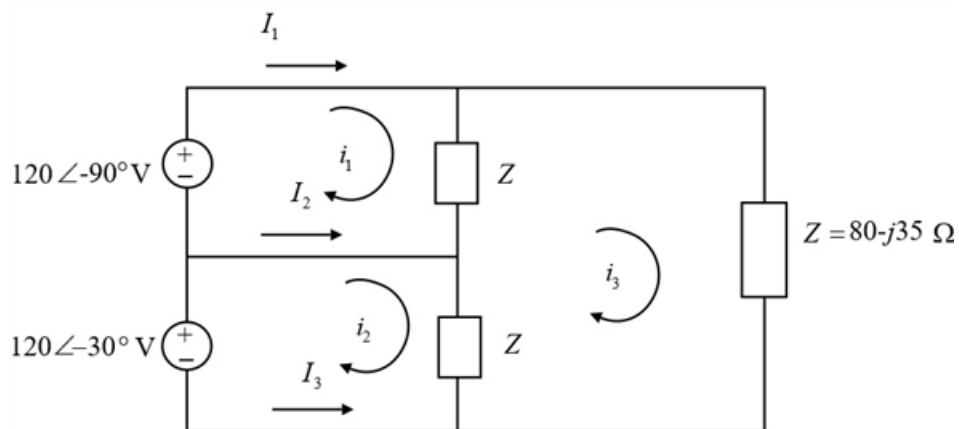


Figure 1

Step 2 of 4

Apply Kirchhoff's voltage law in loop 1.

$$120\angle -90^\circ - Z_{i_1} + Z_{i_3} = 0$$

$$-j120 = (80 - j35)(i_1 - i_3)$$

$$-j120 = (80 - j35)i_1 - (80 - j35)i_3$$

$$(80 - j35)i_1 - (80 - j35)i_3 = -j120 \dots\dots (1)$$

Apply Kirchhoff's voltage law in loop 2.

$$-120\angle -30^\circ - Z_{i_2} + Z_{i_3} = 0$$

$$-120\angle -30^\circ = (80 - j35)(i_2 - i_3)$$

$$(80 - j35)i_2 - (80 - j35)i_3 = -103.92 + j60 \dots\dots (2)$$

Apply Kirchhoff's voltage law in loop 3.

$$-Z_{i_3} + Z_{i_2} - Z_{i_3} + Z_{i_1} - Z_{i_3} = 0$$

$$-Z_{i_1} - Z_{i_2} + 3Z_{i_3} = 0$$

$$-(80 - j35)i_1 - (80 - j35)i_2 + 3(80 - j35)i_3 = 0 \dots\dots (3)$$

Step 3 of 4

Write the MATLAB code for the Kirchhoff's voltage law equations.

```
A=[80-35i 0 -80+35i; 0 80-35i -80+35i; -80+35i
```

```
-80+35i 240-105i];
```

```
B=[-120i; -103.92+60i; 0];
```

```
I=inv(A)*B
```

The MATLAB output is,

I =

-0.2641 - 2.3655i

-2.1806 - 0.9540i

-0.8149 - 1.1065i

Step 4 of 4

Calculate the currents.

$$I_1 = i_1$$

$$= -0.2641 - j2.3655$$

$$= 2.38\angle -96.37^\circ \text{ A}$$

$$I_2 = i_2 - i_1$$

$$= -2.1806 - j0.9540 - (-0.2641 - j2.3655)$$

$$= -1.9165 + j1.4115$$

$$= 2.38\angle 143.63^\circ \text{ A}$$

$$I_3 = -i_2$$

$$= -(-2.1806 - j0.9540)$$

$$= 2.38\angle 23.63^\circ \text{ A}$$

Therefore, the currents are

$$I_1 = 2.38\angle -96.37^\circ \text{ A}, I_2 = 2.38\angle 143.63^\circ \text{ A}, I_3 = 2.38\angle 23.63^\circ \text{ A}.$$